Kai Gao

Contact Information

kgao33@wisc.edu | kgao33.github.io
(608) 556-8974
Department of Economics, University of Wisconsin–Madison

Education

University of Wisconsin–Madison, Ph.D. in Economics 2027 (expected)
Fields: Industrial Organization, Environmental Economics
Committee: Panle Jia Barwick (co-primary), Jean-François Houde (co-primary), Karam Kang

University of Wisconsin–Madison, M.S. in Economics 2021

London School of Economics and Political Science, MSc in Econometrics and Mathematical Economics 2018

University of Minnesota, Twin Cities, B.A. in Mathematics 2017

Working Papers

Beyond the Label: Do Ecodesign Regulations Reduce Real-World Energy Use? **(JMP)**

I study the 2021 European washing-machine energy-label reform. Firms improve Eco efficiency while energy use rises in other wash programs. An equilibrium model of product design, purchases, and program choice quantifies the implications for actual electricity use and welfare.

What Drives the Repowering of Wind Turbines? Evidence from the Danish Wind Energy Industry

I study how renewable-energy subsidies affect the replacement of aging wind turbines in Denmark. A dynamic model of investment and replacement examines how support for existing installations shapes the timing of technology upgrades and evaluates alternative subsidy designs.

Work in Progress

Fragmented Environmental Regulation in Global Shipping

Instrument Choice in Merger Simulations
with Adam Dearing, Lorenzo Magnolfi, and Christopher Sullivan

Work Experience

Research Assistant to Lorenzo Magnolfi 2020–2022

Fellowships and Awards

UW–Madison Data Funding 2026
Robock Scholarship in Empirical or Experimental Economics 2025
Julian Granger Outstanding TA Scholarship 2023
Harold Hardy Scholarship in Economics, University of Minnesota 2016

Conferences and Seminars

- Industrial Organization Workshop, University of Wisconsin–Madison 2026
- Industrial Organization Lunch, University of Wisconsin–Madison 2026
- International Industrial Organization Conference (IIOC) 2025
- Alumni Conference, University of Wisconsin–Madison 2025

Teaching Experience

Teaching Assistant, Econ 310: Statistics: Measurement in Economics, University of Wisconsin–Madison — Spring, Fall 2026

Instructor, Econ 460: Economic Forecasting, University of Wisconsin–Madison — Fall 2025

Head Teaching Assistant, Econ 101: Principles of Microeconomics, University of Wisconsin–Madison — Spring 2025

Teaching Assistant, Master's Econometrics, University of Wisconsin–Madison — Fall 2023

Teaching Assistant, Ph.D. Math Camp, University of Wisconsin–Madison — Summer 2022

References

Panle Jia Barwick (co-primary)
Professor; Todd E. and Elizabeth H. Warnock Distinguished Chair
Department of Economics, University of Wisconsin–Madison
pbarwick@wisc.edu
Phone: +1 (608) 264-2973

Jean-François Houde (co-primary)
Professor; David Edwin and Lucille Hartmann Davies Chair in Economics
Department of Economics, University of Wisconsin–Madison
houde@wisc.edu
Phone: +1 (608) 262-7927

Karam Kang
Associate Professor; Laurits R. Christensen Professor in Economics
Department of Economics, University of Wisconsin–Madison
kkang57@wisc.edu
Phone: +1 (608) 265-6258